Andrew Zhao

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Positions

Sandia National Laboratories (Livermore, CA) 2026–present
Senior Member of Technical Staff

Sandia National Laboratories (Livermore, CA) 2024–2026
Gil Herrera Postdoctoral Fellow in Quantum Information Science

Education

PhD in Physics, University of New Mexico 2018–2024
Advisor: Akimasa Miyake
Dissertation: *Learning, Optimizing, and Simulating Fermions with Quantum Computers*
Chairman's Award for Best Dissertation, Department of Physics and Astronomy, 2024

BSc in Physics, University of Maryland, College Park 2013–2018

Research Experience

Google Quantum AI (San Francisco, CA) 2022
Student Researcher
Host: Nicholas Rubin

Joint Center for Quantum Information and Computer Science (University of Maryland, College Park) 2016–2017
Undergraduate Research Assistant
Mentor: Shelby Kimmel

Grants

"Investigating quantum algorithms and complexity for applications to physical simulation" 2024–2026
US$265,000
Project PI. Funded by the Laboratory Directed R&D program at Sandia National Laboratories, Computing & Information Sciences Division, as part of the Gil Herrera Fellowship. Supported self (postdoc) and two graduate student interns.

Publications

- E. J. Bobrow, R. W. Chien, A. Kononov, J. S. Nelson, **A. Zhao**, L. K. Kovalsky, A. D. Baczewski, "Thermal state preparation for quantum simulations of warm dense matter," in preparation, 2026.
- A. Christensen, **A. Zhao**, "Learning fermionic linear optics with Heisenberg scaling and physical operations," arXiv:2602.05058, 2026.
- B. Harrison, V. Iyer, O. Parekh, K. Thompson, **A. Zhao**, "Fermionic Insights into Measurement-Based Quantum Computation: Circle Graph States Are Not Universal Resources," arXiv:2510.05557, 2025. *Accepted at TQC 2026, non-proceedings track.*
- **A. Zhao**, "Learning the Structure of Any Hamiltonian from Minimal Assumptions," *Proceedings of the 57th Annual ACM Symposium on Theory of Computing (STOC)*, 1201, 2025. *Also accepted at the QIP 2025.*
- **A. Zhao**, N. C. Rubin, "Expanding the reach of quantum optimization with fermionic embeddings," *Quantum* **8**, 1451, 2024.
- **A. Zhao**, A. Miyake, "Group-theoretic error mitigation enabled by classical shadows and symmetries," *npj Quantum Information* **10**, 57, 2024.
- R. Babbush, W. J. Huggins, D. W. Berry, S. F. Ung, **A. Zhao**, D. R. Reichman, H. Neven, A. D. Baczewski, J. Lee, "Quantum simulation of exact electron dynamics can be more efficient than classical mean-field methods," *Nature Communications* **14**, 4058, 2023.
- A. K. Daniel, Y. Zhu, C. H. Alderete, V. Buchemmavari, A. M. Green, N. H. Nguyen, T. G. Thurtell, **A. Zhao**, N. M. Linke, A. Miyake, "Quantum computational advantage attested by nonlocal games with the cyclic cluster state," *Physical Review Research* **4**, 033068, 2022.
- **A. Zhao**, N. C. Rubin, A. Miyake, "Fermionic Partial Tomography via Classical Shadows," *Physical Review Letters* **127**, 110504, 2021.
- **A. Zhao**, A. Tranter, W. M. Kirby, S. F. Ung, A. Miyake, P. J. Love, "Measurement reduction in variational quantum algorithms," *Physical Review A* **101**, 062322, 2020.

Patents

- N. C. Rubin, **A. Zhao**, "Solving quadratic optimization problems over orthogonal groups using a quantum computer," US Patent US 2024 0296367 A1, 2024.

Talks

Invited

- "Learning the structure of any Hamiltonian from minimal assumptions" — Feb 2026
  Visa Research Quantum Seminar
- "Quantum learning theory as a tool for faster quantum simulation" — Feb 2026
  Sandia Quantum Information Science Seminar
- "Hamiltonian learning: A survey of recent progress" — Mar 2025
  University of Sherbrooke Institut quantique Colloquium
- "Group-theoretic error mitigation enabled by classical shadows and symmetries" — Nov 2023
  University of Oxford Quantum Technology Theory Seminar
- "Quantum relaxation for quadratic programs over orthogonal matrices" — Oct 2023
  INFORMS Annual Meeting
- "Efficient and robust learning of quantum many-body properties with classical shadows" — Jun 2023
  Sandia Quantum Computer Science Seminar
- "Efficient and robust learning of fermionic reduced density matrices with classical shadows" — Jan 2023
  UC Berkeley Quantum Many-Body Seminar

Contributed

- "Learning the structure of any Hamiltonian from minimal assumptions" — Jun 2025
  ACM Symposium on Theory of Computing (STOC)
- "Learning the structure of any Hamiltonian from minimal assumptions" — Mar 2025
  APS Global Physics Summit
- "Learning the structure of any Hamiltonian from minimal assumptions" — Feb 2025
  Quantum Information Processing (QIP) Conference
- "Nearly Heisenberg-limited Hamiltonian learning with few assumptions" — Oct 2024
  Southwest Quantum Information and Technology (SQuInT) Workshop
- "Group-theoretic error mitigation enabled by classical shadows and symmetries" — Oct 2023
  Southwest Quantum Information and Technology (SQuInT) Workshop
- "Calibration-free quantum error mitigation with classical shadows" — Mar 2023
  APS March Meeting
- "Fermionic partial tomography via classical shadows" — Mar 2021
  APS March Meeting
- "Measurement reduction in variational quantum algorithms" — Mar 2020
  APS March Meeting

Teaching

Courses

- General Physics (PHYC 151), Graduate Supplemental Instructor — Spring 2019
  University of New Mexico, Department of Physics and Astronomy
- General Physics Laboratory (PHYC 151L), Graduate Lab Instructor — Fall 2018
  University of New Mexico, Department of Physics and Astronomy

Guest Lectures

- "Classical shadows are both simpler and more complicated than you think" — Mar 2025
  University of Sherbrooke, Institut quantique, seminar
- "Quantum metrology and distributed quantum sensing" — Jun 2023
  University of New Mexico, Center for Quantum Information and Control, summer course
- "Random unitary evolution, t-designs, and applications to quantum chaos" — Jun 2021
  University of New Mexico, Center for Quantum Information and Control, summer course
- "Continuous symmetries and $\mathrm{O}(N)$ models" — Jun 2020
  University of New Mexico, Center for Quantum Information and Control, summer course
- "Simple Lie groups and simple Lie algebras" — Jun 2019
  University of New Mexico, Center for Quantum Information and Control, summer course

Advising

Sandia Graduate Interns

- Ramya Bhaskar, *University of Washington, PhD student* 2024–2025
  – Now a postdoc at Sandia
- Aria Christensen, *Ohio State University, PhD student* 2024–2026

Professional Service

Journal Referee

- Physical Review Letters • Physical Review A • Physical Review Research • PRX Quantum • Journal of Chemical Theory and Computation • Nature Communications • npj Quantum Information • Quantum • Communications in Mathematical Physics

Conference Referee

- QIP • TQC • QSim • QCTiP

Grant Proposal Reviewer

- Sandia LDRD program